Teacher Wu Kuan | Exam at a Glance #24 — Dot Product & Perpendicular Vectors

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1. The trap question

(3, t) and (2, −6) are perpendicular. Find t.

2. The method

- $a \cdot b = |a||b| \cos\theta$; with coordinates $a \cdot b = a_1b_1 + a_2b_2$.
- Perpendicular $\Leftrightarrow a \cdot b = 0$ ($\cos 90° = 0$).
- $3 \cdot 2 + t \cdot (-6) = 0 \rightarrow t = 1$.
- Angle: $\cos\theta = a \cdot b / (|a||b|)$. $|a - b|^2 = |a|^2 + |b|^2 - 2a \cdot b$.

3. Parent check method

Ask: “Is the dot product a number or a vector?” (A number.)

Practice

1. $a = (3, 4)$, $b = (2, -1)$. Find $a \cdot b$.
2. $|a| = 2$, $|b| = 3$, angle 60°. Find $a \cdot b$.
3. $a = (1, 2)$, $b = (k, 3)$ are perpendicular. Find k.
4. Find the angle between $a = (3, 1)$ and $b = (1, 2)$.
5. $|a| = 2$, $|b| = 3$, $|a - b| = \sqrt{7}$. Find the angle between a and b.

Answers

1. 2
2. 3
3. $k = -6$
4. 45°
5. 60°

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